

Research Report: ROS2 Homework 1 & 2

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April 8, 2026

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All the code has been uploaded [here](#).

1 HOMEWORK 1

#	Criteria	Verification Method
1	Package builds without warnings	<code>colcon build</code>
2	<code>distance_publisher</code> publishes at 1 Hz	<code>ros2 topic hz /distance_topic</code>
3	<code>distance_listener</code> prints warning when $\text{distance} < \text{threshold}$	Observe logs
4	Service can update threshold at runtime	<code>ros2 service call</code>
5	Listener uses updated threshold after service call	Observe logs
6	Action server provides all 5 feedback steps	Run action client
7	Action result matches threshold condition correctly	Test multiple values
8	Launch file starts the whole system	<code>ros2 launch ...</code>
9	<code>rqt_graph</code> shows all nodes and topics	<code>rqt_graph</code>

Table 1: Acceptance Criteria for BTVN_01

1. PACKAGE BUILDS WITHOUT WARNINGS

```
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ colcon build --packages-select distance_warning
Starting >>> distance_warning
Finished <<< distance_warning [8.68s]
Summary: 1 package finished [9.42s]
```

colcon built successfully

2. DISTANCE_PUBLISHER PUBLISHES AT 1 HZ

```
^Cngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 topic hz /distance_topic
average rate: 1.001
  min: 0.998s max: 1.000s std dev: 0.00086s window: 3
average rate: 1.001
  min: 0.998s max: 1.001s std dev: 0.00105s window: 4
average rate: 1.000
  min: 0.998s max: 1.001s std dev: 0.00096s window: 6
average rate: 1.000
  min: 0.998s max: 1.001s std dev: 0.00102s window: 7
average rate: 1.000
  min: 0.998s max: 1.001s std dev: 0.00107s window: 9
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ |
```

topic rate

The average rate of the `/distance_topic` is around 1 Hz.

3. DISTANCE_LISTENER PRINTS WARNING WHEN DISTANCE < THRESHOLD

```
ngxwang@LAPTOP-SS0001C1: ~$ ros2 launch distance_publisher distance_publisher.py
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [distance_publisher-1]: process started with pid [177786]
[INFO] [distance_listener-2]: process started with pid [177788]
[INFO] [set_threshold_service-3]: process started with pid [177790]
[INFO] [distance_action_server-4]: process started with pid [177792]
[distance_publisher-1] [INFO] [1775627021.631701315] [distance_publisher]: Distance Publisher Node created
[distance_listener-2] [INFO] [1775627021.632352708] [distance_listener]: Distance Subscriber Node created
[set_threshold_service-3] [INFO] [1775627021.635173780] [set_threshold_service]: Service Set Threshold created!
[distance_publisher-1] [INFO] [1775627022.632167685] [distance_publisher]: Publishing: 0.15 m
[distance_listener-2] [WARN] [1775627022.633089159] [distance_listener]: Warning: Object too close! (0.15 m < threshold: 0.50 m)
[distance_publisher-1] [INFO] [1775627023.631841676] [distance_publisher]: Publishing: 0.69 m
[distance_listener-2] [INFO] [1775627023.632253689] [distance_listener]: Distance received: 0.69 m
[distance_publisher-1] [INFO] [1775627024.632513806] [distance_publisher]: Publishing: 1.18 m
[distance_listener-2] [INFO] [1775627024.632513806] [distance_listener]: Distance received: 1.18 m
[distance_publisher-1] [INFO] [1775627025.631954706] [distance_publisher]: Publishing: 1.40 m
[distance_listener-2] [INFO] [1775627025.632457049] [distance_listener]: Distance received: 1.40 m
[distance_publisher-1] [INFO] [1775627026.631918396] [distance_publisher]: Publishing: 0.11 m
[distance_listener-2] [WARN] [1775627026.632838939] [distance_listener]: Warning: Object too close! (0.11 m < threshold: 0.50 m)
[distance_publisher-1] [INFO] [1775627027.632744463] [distance_publisher]: Publishing: 1.31 m
[distance_listener-2] [INFO] [1775627027.632744463] [distance_listener]: Distance received: 1.31 m
[distance_publisher-1] [INFO] [1775627028.631997620] [distance_publisher]: Publishing: 1.39 m
[distance_listener-2] [INFO] [1775627028.632911701] [distance_listener]: Distance received: 1.39 m
[distance_publisher-1] [INFO] [1775627029.631974898] [distance_publisher]: Publishing: 1.16 m
[distance_listener-2] [INFO] [1775627029.632882156] [distance_listener]: Distance received: 1.16 m
[distance_publisher-1] [INFO] [1775627030.632242597] [distance_publisher]: Publishing: 0.40 m
[distance_listener-2] [WARN] [1775627030.633041062] [distance_listener]: Warning: Object too close! (0.40 m < threshold: 0.50 m)
[distance_publisher-1] [INFO] [1775627031.632131081] [distance_publisher]: Publishing: 0.99 m
[distance_listener-2] [INFO] [1775627031.632793051] [distance_listener]: Distance received: 0.99 m
[distance_publisher-1] [INFO] [1775627032.631994975] [distance_publisher]: Publishing: 1.03 m
[distance_listener-2] [INFO] [1775627032.632691063] [distance_listener]: Distance received: 1.03 m
[distance_publisher-1] [INFO] [1775627033.632103219] [distance_publisher]: Publishing: 0.41 m
[distance_listener-2] [WARN] [1775627033.632647115] [distance_listener]: Warning: Object too close! (0.41 m < threshold: 0.50 m)
[distance_publisher-1] [INFO] [1775627034.631944403] [distance_publisher]: Publishing: 0.95 m
```

Warning when distance < threshold

4. SERVICE CAN UPDATE THRESHOLD AT RUNTIME

```
[distance_publisher-1] [INFO] [1775630709.239051378] [distance_publisher]: Publishing: 0.36 m
[distance_listener-2] [WARN] [1775630709.240047876] [distance_listener]: Warning: Object too close! (0.36 m < threshold: 0.80 m)
[distance_publisher-1] [INFO] [1775630710.238949980] [distance_publisher]: Publishing: 0.69 m
[distance_listener-2] [WARN] [1775630710.239311914] [distance_listener]: Warning: Object too close! (0.69 m < threshold: 0.80 m)
[distance_publisher-1] [INFO] [1775630711.239058781] [distance_publisher]: Publishing: 1.00 m
[distance_listener-2] [INFO] [1775630711.240065326] [distance_listener]: Distance received: 1.00 m
[distance_publisher-1] [INFO] [1775630712.239102872] [distance_publisher]: Publishing: 0.70 m
[distance_listener-2] [WARN] [1775630712.239839939] [distance_listener]: Warning: Object too close! (0.70 m < threshold: 0.70 m)
[distance_publisher-1] [INFO] [1775630713.238957689] [distance_publisher]: Publishing: 0.51 m
[distance_listener-2] [WARN] [1775630713.239448220] [distance_listener]: Warning: Object too close! (0.51 m < threshold: 0.70 m)
[distance_publisher-1] [INFO] [1775630714.239168209] [distance_publisher]: Publishing: 0.77 m

ngxwang@LAPTOP-SS0001C1: ~$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: true}"
requester: making request: distance_warning.srv.SetThreshold_Request(increase=True)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=1.5, message='Threshold increased to: 1.500000 m')

ngxwang@LAPTOP-SS0001C1: ~$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: true}"
requester: making request: distance_warning.srv.SetThreshold_Request(increase=True)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=1.5, message='Threshold increased to: 1.500000 m')

ngxwang@LAPTOP-SS0001C1: ~$ ros2 param set /distance_listener threshold 0.8
Set parameter successful
ngxwang@LAPTOP-SS0001C1: ~$ ros2 param set /distance_listener threshold 0.7
Set parameter successful
ngxwang@LAPTOP-SS0001C1: ~$
```

Update the listener's threshold directly using `param set`

```

ng: 0.98 m
[distance_listener-2] [INFO] [1775631734.942811114] [distance_listener]: Distance received: 0.98 m
[set_threshold_service-3] [INFO] [1775631735.718372570] [set_threshold_service]: Threshold updated: 0.10 -> 0.20
[set_threshold_service-3] [INFO] [1775631735.718648420] [set_threshold_service]: Updated listener threshold: 0.20
[distance_publisher-1] [INFO] [1775631735.942097907] [distance_publisher]: Publishing: 0.72 m
[distance_listener-2] [INFO] [1775631735.942990711] [distance_listener]: Distance received: 0.72 m
[distance_publisher-1] [INFO] [1775631736.942109356] [distance_publisher]: Publishing: 0.73 m
[distance_listener-2] [INFO] [1775631736.942712769] [distance_listener]: Distance received: 0.73 m

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.1000000149011612, message='Threshold increased to: 0.100000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws/src$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: false}"
requester: making request: distance_warning.srv.SetThreshold_Request(increase=False)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.1000000149011612, message='Threshold increased to: 0.100000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws/src$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: true}"
waiting for service to become available...
requester: making request: distance_warning.srv.SetThreshold_Request(increase=True)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.2000000298023224, message='Threshold increased to: 0.200000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws/src$

```

Update the listener's threshold through service call

Service updates the threshold after the call:

ros2 service call /set_threshold distance_warning/srv/SetThreshold "increase: true"
(0.1 → 0.2)

5. LISTENER USES UPDATED THRESHOLD AFTER SERVICE CALL

```

[INFO] [1775632277.086461716] [distance_publisher]: Publishing: 0.17 m
[INFO] [1775632278.086477738] [distance_publisher]: Publishing: 0.63 m
[INFO] [1775632279.086230586] [distance_publisher]: Publishing: 0.61 m
[INFO] [1775632280.086230451] [distance_publisher]: Publishing: 0.95 m
[INFO] [1775632281.086237675] [distance_publisher]: Publishing: 1.40 m
[INFO] [1775632282.08629562] [distance_publisher]: Publishing: 1.20 m
[INFO] [1775632283.086362286] [distance_publisher]: Publishing: 0.44 m
[INFO] [1775632284.086343917] [distance_publisher]: Publishing: 1.21 m
[INFO] [1775632285.086326651] [distance_publisher]: Publishing: 1.29 m
[INFO] [1775632286.091841321] [distance_publisher]: Publishing: 0.68 m
[INFO] [1775632287.086308000] [distance_publisher]: Publishing: 1.21 m
[INFO] [1775632288.086373585] [distance_publisher]: Publishing: 0.68 m
[INFO] [1775632289.086661286] [distance_publisher]: Publishing: 1.12 m
[INFO] [1775632290.086462417] [distance_publisher]: Publishing: 0.17 m
[INFO] [1775632291.086254089] [distance_publisher]: Publishing: 0.66 m
[INFO] [1775632292.086349460] [distance_publisher]: Publishing: 1.21 m
[INFO] [1775632293.086268192] [distance_publisher]: Publishing: 0.50 m
[INFO] [1775632294.086319756] [distance_publisher]: Publishing: 0.30 m
[INFO] [1775632295.086356992] [distance_publisher]: Publishing: 0.79 m
[INFO] [1775632296.086258852] [distance_publisher]: Publishing: 0.77 m

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.5, message='Threshold increased to: 0.500000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: false}"
requester: making request: distance_warning.srv.SetThreshold_Request(increase=False)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.4000000059604645, message='Threshold increased to: 0.400000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: false}"
requester: making request: distance_warning.srv.SetThreshold_Request(increase=False)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.4000000059604645, message='Threshold increased to: 0.400000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws$

[INFO] [1775632280.086977930] [distance_listener]: Distance Subscriber Mode created
[INFO] [1775632280.086952726] [distance_listener]: Distance received: 0.95 m
[INFO] [1775632281.086966882] [distance_listener]: Distance received: 1.40 m
[INFO] [1775632282.086991798] [distance_listener]: Distance received: 1.20 m
[WARN] [1775632283.087177951] [distance_listener]: Warning: Object too close! (0.44 m < threshold: 0.50 m)
[INFO] [1775632284.086977909] [distance_listener]: Distance received: 1.21 m
[INFO] [1775632285.086973961] [distance_listener]: Distance received: 1.29 m
[INFO] [1775632286.09588397] [distance_listener]: Distance received: 0.68 m
[INFO] [1775632287.086939263] [distance_listener]: Distance received: 1.21 m
[INFO] [1775632288.087139643] [distance_listener]: Distance received: 0.68 m
[INFO] [1775632289.087204825] [distance_listener]: Distance received: 1.12 m
[WARN] [1775632290.086976968] [distance_listener]: Warning: Object too close! (0.17 m < threshold: 0.40 m)
[INFO] [1775632291.086853888] [distance_listener]: Distance received: 0.66 m
[INFO] [1775632292.087027123] [distance_listener]: Distance received: 1.21 m
[INFO] [1775632293.086810854] [distance_listener]: Distance received: 0.50 m
[WARN] [1775632294.086926448] [distance_listener]: Warning: Object too close! (0.30 m < threshold: 0.40 m)
[INFO] [1775632295.086927611] [distance_listener]: Distance received: 0.79 m
[INFO] [1775632296.086608596] [distance_listener]: Distance received: 0.77 m

[INFO] [1775631715.293521930] [set_threshold_service]: Threshold updated: 1.00 -> 0.90
[INFO] [1775631718.996181737] [set_threshold_service]: Threshold updated: 0.90 -> 0.80
[INFO] [1775631722.677828297] [set_threshold_service]: Threshold updated: 0.80 -> 0.70
[INFO] [1775631726.625533920] [set_threshold_service]: Threshold updated: 0.70 -> 0.60
[INFO] [1775631829.733597268] [set_threshold_service]: Threshold updated: 0.60 -> 0.50
[INFO] [1775632188.695918298] [set_threshold_service]: Threshold updated: 0.70 -> 0.80
[INFO] [1775632213.765899083] [set_threshold_service]: Threshold updated: 0.80 -> 0.70
[INFO] [1775632218.734703437] [set_threshold_service]: Threshold updated: 0.70 -> 0.60
[INFO] [1775632222.302704204] [set_threshold_service]: Threshold updated: 0.60 -> 0.50
[INFO] [1775632227.956723541] [set_threshold_service]: Threshold updated: 0.50 -> 0.40
[CINFO] [1775632269.160276532] [rclcpp]: signal handler(SIGINT/SIGTERM)
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 colcon build --packages-select distance_warning
Starting >>> distance_warning
Finished <<< distance_warning [0.43s]

Summary: 1 package finished [0.69s]
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 run distance_warning set_threshold_service
[INFO] [1775632277.616523193] [set_threshold_service]: Service Set Threshold created
[INFO] [1775632286.616559223] [set_threshold_service]: Threshold updated: 0.50 -> 0.40
[INFO] [1775632286.616853682] [set_threshold_service]: Updated listener threshold: 0.40

```

Threshold updated successfully through service call

When the service is called, it updates the threshold both in the service and listener ($0.5 \rightarrow 0.4$).

6. ACTION SERVER PROVIDES ALL 5 FEEDBACK STEPS

```
ngxwang@LAPTOP-S50001C1:~/ros2_ws$ ros2 action send_goal --feedback /distance_action distance_warning/action/Distance "{distance_to_check: 0.4}"
The passed action type is invalid
ngxwang@LAPTOP-S50001C1:~/ros2_ws$ ros2 action send_goal --feedback /check_distance distance_warning/action/CheckDistance "{distance_to_check: 0.4}"
Waiting for an action server to become available...
Sending goal:
  distance_to_check: 0.4

Goal accepted with ID: 7e8182f68daa42c2ac7afb7c414dad71

Feedback:
  step: 1
  total_steps: 5
  feedback_msg: Receiving distance value...

Feedback:
  step: 2
  total_steps: 5
  feedback_msg: Fetching threshold parameter...

Feedback:
  step: 3
  total_steps: 5
  feedback_msg: Comparing values...

Feedback:
  step: 4
  total_steps: 5
  feedback_msg: Generating result...

Feedback:
  step: 5
  total_steps: 5
  feedback_msg: Done.

Result:
  is_safe: false
  result_message: 'Warning: Too close to obstacle!'

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-S50001C1:~/ros2_ws$
```

5 feedback steps

7. ACTION RESULT MATCHES THRESHOLD CONDITION CORRECTLY

```
The passed action type is invalid
ngxwang@LAPTOP-S50001C1:~/ros2_ws$ ros2 action send_goal /check_distance distance_warning/a
ction/CheckDistance "{distance_to_check: 0.8}"
Waiting for an action server to become available...
Sending goal:
  distance_to_check: 0.8

Goal accepted with ID: fcd07cd6781e4451a63b11df797cb837

Result:
  is_safe: true
  result_message: Safe distance!

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-S50001C1:~/ros2_ws$ ros2 action send_goal /check_distance distance_warning/action/CheckDistance "{distance_to_check: 0.4}"
Waiting for an action server to become available...
Sending goal:
  distance_to_check: 0.4

Goal accepted with ID: 7c99c9be9ab840f3975fcee43519f500

Result:
  is_safe: false
  result_message: 'Warning: Too close to obstacle!'

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-S50001C1:~/ros2_ws$ ros2 action send_goal /check_distance distance_warning/action/CheckDistance "{distance_to_check: 0.5}"
Waiting for an action server to become available...
Sending goal:
  distance_to_check: 0.5

Goal accepted with ID: 28589c0badf744c7a22dc354dfcd704a

Result:
  is_safe: false
  result_message: 'Warning: Too close to obstacle!'

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-S50001C1:~/ros2_ws$
```

threshold = 0.6

```

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 service call /set_threshold distance_warning/srv/SetThreshold "{increase: true}"
waiting for service to become available...
requester: making request: distance_warning.srv.SetThreshold_Request(increase=True)

response:
distance_warning.srv.SetThreshold_Response(success=True, new_threshold=0.699999988879071, message='Threshold increased to: 0.700000 m')

ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 action send_goal /check_distance distance_warning/action/CheckDistance "{distance_to_check: 0.65}"
Waiting for an action server to become available...
Sending goal:
  distance_to_check: 0.65

Goal accepted with ID: 0747a8e539c04428b3bcd652758b3d65

Result:
  is_safe: false
  result_message: 'Warning: Too close to obstacle!'

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 action send_goal /check_distance distance_warning/action/CheckDistance "{distance_to_check: 0.75}"
Waiting for an action server to become available...
Sending goal:
  distance_to_check: 0.75

Goal accepted with ID: d211209b35ee4c3d9acd4cb62674171f

Result:
  is_safe: true
  result_message: 'Safe distance!'

Goal finished with status: SUCCEEDED
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ |

```

threshold = 0.7

8. LAUNCH FILE STARTS THE WHOLE SYSTEM

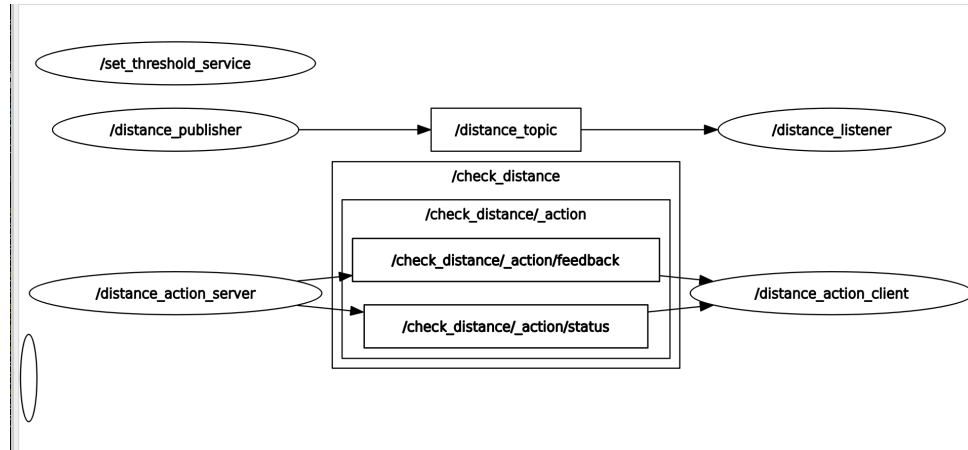
```

ngxwang@LAPTOP-SS0001C1: ~
[WARN] [1775635277.162397106] [distance_listener]: Warning: Object too close! (0.41 m < threshold: 0.50 m)
[INFO] [1775635278.162152362] [distance_listener]: Distance received: 1.07 m
^C[INFO] [1775635278.551487737] [rclcpp]: signal_handler(SIGINT/SIGTERM)
ngxwang@LAPTOP-SS0001C1:~/ros2_ws/src$ ros2 launch distance_warning distance_warning.launch.py
[INFO] [launch]: All log files can be found below /home/ngxwang/.ros/log/2026-04-08-15-01-40-276428-LAPTOP-SS0001C1-21446
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [distance_publisher-1]: process started with pid [21453]
[INFO] [distance_listener-2]: process started with pid [21455]
[INFO] [set_threshold_service-3]: process started with pid [21457]
[INFO] [distance_action_server-4]: process started with pid [21459]
[distance_publisher-1] [INFO] [1775635300.433233091] [distance_publisher]: Distance Publisher Node created
[distance_listener-2] [INFO] [1775635300.433893577] [distance_listener]: Distance Subscriber Node created
[set_threshold_service-3] [INFO] [1775635300.441093135] [set_threshold_service]: Service Set Threshold created!
[distance_listener-2] [INFO] [1775635301.162450245] [distance_listener]: Distance received: 0.92 m
[distance_publisher-1] [INFO] [1775635301.433441923] [distance_publisher]: Publishing: 1.32 m
[distance_listener-2] [INFO] [1775635301.434040415] [distance_listener]: Distance received: 1.32 m
[distance_listener-2] [INFO] [1775635302.161669929] [distance_listener]: Distance received: 0.67 m
[distance_publisher-1] [INFO] [1775635302.433785831] [distance_publisher]: Publishing: 0.42 m
[distance_listener-2] [WARN] [1775635302.434855224] [distance_listener]: Warning: Object too close! (0.42 m < threshold: 0.50 m)
[distance_listener-2] [INFO] [1775635303.161658845] [distance_listener]: Distance received: 1.47 m
[distance_publisher-1] [INFO] [1775635303.43387220] [distance_publisher]: Publishing: 0.92 m
[distance_listener-2] [INFO] [1775635303.433787612] [distance_listener]: Distance received: 0.92 m
[distance_publisher-1] [INFO] [1775635304.162237682] [distance_publisher]: Publishing: 1.41 m
[distance_listener-2] [INFO] [1775635304.433432882] [distance_listener]: Distance received: 1.18 m
[distance_publisher-1] [INFO] [1775635304.433966182] [distance_publisher]: Publishing: 1.18 m
[distance_listener-2] [INFO] [1775635305.162465403] [distance_listener]: Distance received: 1.15 m
[distance_publisher-1] [INFO] [1775635305.433372289] [distance_publisher]: Publishing: 1.33 m
[distance_listener-2] [INFO] [1775635305.433745863] [distance_listener]: Distance received: 1.33 m
[distance_publisher-1] [INFO] [1775635306.161530612] [distance_publisher]: Publishing: 0.61 m
[distance_listener-2] [INFO] [1775635306.433390914] [distance_listener]: Distance received: 1.17 m
[distance_publisher-1] [INFO] [1775635306.433976929] [distance_publisher]: Publishing: 1.17 m
[distance_listener-2] [INFO] [1775635307.162426080] [distance_listener]: Distance received: 0.57 m
[distance_publisher-1] [INFO] [1775635307.433424240] [distance_publisher]: Publishing: 0.50 m
[distance_listener-2] [WARN] [1775635307.434421474] [distance_listener]: Warning: Object too close! (0.50 m < threshold: 0.50 m)
[distance_listener-2] [WARN] [1775635308.161761409] [distance_listener]: Warning: Object too close! (0.29 m < threshold: 0.50 m)

```

Running launch file

9. RQT_GRAPH SHOWS ALL NODES AND TOPICS



rqt_graph

2 HOMEWORK 2

#	Criteria	Verification Method
1	Package builds without warnings after adding dependencies	colcon build
2	distance_tf_broadcaster publishes both frames to /tf	ros2 run tf2_ros tf2_echo base_link sensor_link
3	sensor_link x changes according to /distance_topic	Compare logs
4	distance_tf_listener computes correct distance	Observe logs
5	When sensor exceeds tf_threshold, service /set_threshold is triggered	Observe logs of both nodes
6	/distance_reliable publishes at 1 Hz	ros2 topic hz /distance_reliable
7	/distance_best_effort publishes at 10 Hz	ros2 topic hz /distance_best_effort
8	Statistics are printed correctly every 5 seconds	Observe logs
9	QoS incompatibility warning appears	ros2 topic echo ... -qos-reliability reliable
10	Full launch file starts all nodes	rqt_graph

Table 2: Acceptance Criteria for BTVN_02

1. PACKAGE BUILDS WITHOUT WARNINGS AFTER ADDING DEPENDENCIES


```
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ colcon build --packages-select distance_warning
Starting >>> distance_warning
Finished <<< distance_warning [4.74s]

Summary: 1 package finished [5.49s]
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$
```

colcon built successfully

2. DISTANCE_TF_BROADCASTER PUBLISHES BOTH FRAMES TO /TF

```
0.000 0.000 0.000 1.000
At time 1775636455.796133080
- Translation: [1.426, 0.000, 0.000]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.000, 1.000]
- Rotation: in RPY (radian) [0.000, -0.000, 0.000]
- Rotation: in RPY (degree) [0.000, -0.000, 0.000]
- Matrix:
1.000 0.000 0.000 1.426
0.000 1.000 0.000 0.000
0.000 0.000 1.000 0.000
0.000 0.000 0.000 1.000
At time 1775636456.796244858
- Translation: [1.146, 0.000, 0.000]
- Rotation: in Quaternion (xyzw) [0.000, 0.000, 0.000, 1.000]
- Rotation: in RPY (radian) [0.000, -0.000, 0.000]
- Rotation: in RPY (degree) [0.000, -0.000, 0.000]
- Matrix:
1.000 0.000 0.000 1.146
0.000 1.000 0.000 0.000
0.000 0.000 1.000 0.000
0.000 0.000 0.000 1.000

security Various security related sub-commands
service Various service related sub-commands
topic Various topic related sub-commands
wtf Use 'wtf' as alias to 'doctor'

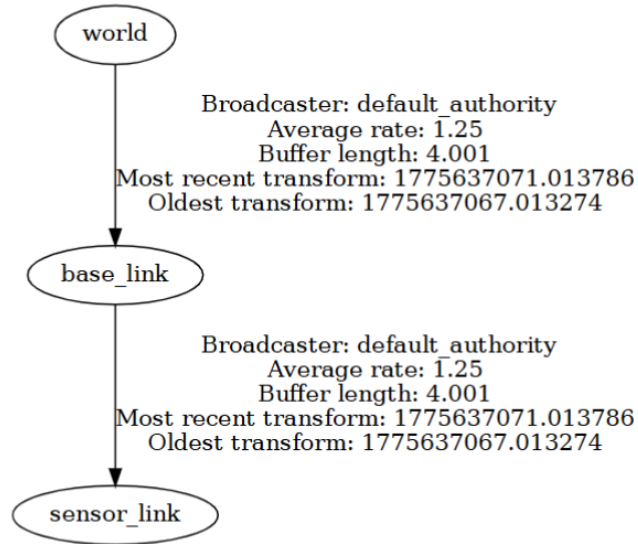
Call 'ros2 <command> -h' for more detailed usage.
ngxwang@LAPTOP-SS0001C1:~/ros2_ws$ ros2 run distance_warning distance_publisher
[INFO] [1775636444.795240749] [distance_publisher]: Distance Publisher Node created
[INFO] [1775636445.795408911] [distance_publisher]: Publishing: 1.16 m
[INFO] [1775636446.795451533] [distance_publisher]: Publishing: 0.23 m
[INFO] [1775636447.795442564] [distance_publisher]: Publishing: 0.44 m
[INFO] [1775636448.795401158] [distance_publisher]: Publishing: 1.19 m
[INFO] [1775636449.795473204] [distance_publisher]: Publishing: 1.01 m
[INFO] [1775636450.795426992] [distance_publisher]: Publishing: 0.97 m
[INFO] [1775636451.795504486] [distance_publisher]: Publishing: 0.30 m
[INFO] [1775636452.795506755] [distance_publisher]: Publishing: 1.26 m
[INFO] [1775636453.795418659] [distance_publisher]: Publishing: 1.02 m
[INFO] [1775636454.795412491] [distance_publisher]: Publishing: 0.82 m
[INFO] [1775636455.795486157] [distance_publisher]: Publishing: 1.43 m
[INFO] [1775636456.795608699] [distance_publisher]: Publishing: 1.15 m
[INFO] [1775636457.795370886] [distance_publisher]: Publishing: 0.58 m

[INFO] [1775636448.795921458] [distance_tf_broadcaster]: Broadcasting sensor_link at x=1.19
[INFO] [1775636449.796135436] [distance_tf_broadcaster]: Broadcasting sensor_link at x=1.01
[INFO] [1775636450.795811926] [distance_tf_broadcaster]: Broadcasting sensor_link at x=0.97
[INFO] [1775636451.796467274] [distance_tf_broadcaster]: Broadcasting sensor_link at x=0.30
[INFO] [1775636452.796336858] [distance_tf_broadcaster]: Broadcasting sensor_link at x=1.26
[INFO] [1775636453.796189848] [distance_tf_broadcaster]: Broadcasting sensor_link at x=1.02
[INFO] [1775636454.795835504] [distance_tf_broadcaster]: Broadcasting sensor_link at x=0.82
[INFO] [1775636455.796239658] [distance_tf_broadcaster]: Broadcasting sensor_link at x=1.43
[INFO] [1775636456.796394339] [distance_tf_broadcaster]: Broadcasting sensor_link at x=1.15
[INFO] [1775636457.795828502] [distance_tf_broadcaster]: Broadcasting sensor_link at x=0.58
```

Broadcaster

view_frames Result

Recorded at time: 1775637071.6281798



view_frames result

3. SENSOR_LINK X CHANGES ACCORDING TO /DISTANCE_TOPIC

```

[INFO] [1775637933.012633983] [distance_publisher]: Publishing: 0.23 m
[INFO] [1775637934.012685992] [distance_publisher]: Publishing: 0.26 m
[INFO] [1775637935.012835598] [distance_publisher]: Publishing: 0.37 m
[INFO] [1775637936.012654413] [distance_publisher]: Publishing: 0.32 m
[INFO] [1775637937.012683359] [distance_publisher]: Publishing: 0.90 m
[INFO] [1775637938.012642053] [distance_publisher]: Publishing: 0.52 m
[INFO] [1775637939.012649969] [distance_publisher]: Publishing: 0.31 m
[INFO] [1775637940.012670330] [distance_publisher]: Publishing: 1.43 m
[INFO] [1775637941.012790567] [distance_publisher]: Publishing: 0.10 m
[INFO] [1775637942.012622935] [distance_publisher]: Publishing: 0.61 m
[INFO] [1775637943.012666884] [distance_publisher]: Publishing: 1.11 m
[INFO] [1775637944.012603528] [distance_publisher]: Publishing: 0.81 m
[INFO] [1775637945.012712355] [distance_publisher]: Publishing: 0.34 m
[INFO] [1775637946.012791583] [distance_publisher]: Publishing: 0.17 m
[INFO] [1775637947.012745598] [distance_publisher]: Publishing: 0.65 m
[INFO] [1775637948.012643692] [distance_publisher]: Publishing: 0.33 m
[INFO] [1775637949.012678910] [distance_publisher]: Publishing: 0.78 m
[INFO] [1775637950.012692684] [distance_publisher]: Publishing: 0.84 m
[INFO] [1775637951.012796234] [distance_publisher]: Publishing: 1.25 m
[INFO] [1775637952.012650347] [distance_publisher]: Publishing: 0.50 m
[INFO] [1775637953.012668735] [distance_publisher]: Publishing: 0.55 m
[INFO] [1775637954.012767788] [distance_publisher]: Publishing: 1.15 m
[INFO] [1775637955.012652270] [distance_publisher]: Publishing: 1.21 m
[INFO] [1775637956.012638164] [distance_publisher]: Publishing: 1.24 m
[INFO] [1775637957.012665221] [distance_publisher]: Publishing: 1.13 m
[INFO] [1775637958.012688382] [distance_publisher]: Publishing: 0.78 m
[INFO] [1775637959.012644416] [distance_publisher]: Publishing: 0.45 m
[INFO] [1775637960.012651114] [distance_publisher]: Publishing: 0.13 m
[INFO] [1775637961.012633937] [distance_publisher]: Publishing: 1.43 m
[INFO] [1775637962.012656446] [distance_publisher]: Publishing: 1.08 m
  
```

x changes according to /distance_topic

Every time publisher publishes another distance value, x changes.

4. DISTANCE_TF_LISTENER COMPUTES CORRECT DISTANCE OBSERVE LOGS


```
ngxwang@LAPTOP-SS0001C1: X + v
min: 0.999s max: 1.001s std dev: 0.00061s window: 27
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00059s window: 29
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00057s window: 31
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00058s window: 32
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00058s window: 34
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00062s window: 36
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00064s window: 38
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00064s window: 40
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00063s window: 41
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00062s window: 43
average rate: 1.000
min: 0.999s max: 1.001s std dev: 0.00063s window: 44
```

/distance_reliable publishes at rate 1Hz

7. /DISTANCE_BEST_EFFORT PUBLISHES AT 10 Hz

```
ngxwang@LAPTOP-SS0001C1: X ngxwang@LAPTOP-SS0001C1: X + v
average rate: 10.000
min: 0.099s max: 0.101s std dev: 0.00042s window: 210
average rate: 10.000
min: 0.099s max: 0.101s std dev: 0.00044s window: 220
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00046s window: 231
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00045s window: 241
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00045s window: 251
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00044s window: 262
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00043s window: 272
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00043s window: 282
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00043s window: 293
average rate: 10.000
min: 0.099s max: 0.102s std dev: 0.00042s window: 303
```

/distance_best_effect publishes at rate 10Hz

8. STATISTICS ARE PRINTED CORRECTLY EVERY 5 SECONDS

```

[INFO] [1775641503.312875562] [distance_listener_qos]: BEST_EFFORT : 50 msg (expected ~50)
[INFO] [1775641503.383070784] [distance_listener_qos]: [BEST_EFFORT] 0.38 m (total: 1)
[INFO] [1775641503.483206850] [distance_listener_qos]: [BEST_EFFORT] 1.28 m (total: 2)
[INFO] [1775641503.583092762] [distance_listener_qos]: [BEST_EFFORT] 0.19 m (total: 3)
[INFO] [1775641503.683302743] [distance_listener_qos]: [BEST_EFFORT] 0.25 m (total: 4)
[INFO] [1775641503.783179556] [distance_listener_qos]: [BEST_EFFORT] 0.17 m (total: 5)
[INFO] [1775641503.883439150] [distance_listener_qos]: [BEST_EFFORT] 0.91 m (total: 6)
[INFO] [1775641503.983331103] [distance_listener_qos]: [BEST_EFFORT] 1.38 m (total: 7)
[INFO] [1775641504.083095695] [distance_listener_qos]: [RELIABLE] 0.23 m (total: 1)
[WARN] [1775641504.083180995] [distance_listener_qos]: [distance_listener_qos]: Warning: Object too close! (0.23 m < threshold: 0.50 m)
[INFO] [1775641504.083237328] [distance_listener_qos]: [BEST_EFFORT] 0.21 m (total: 8)
[INFO] [1775641504.183192774] [distance_listener_qos]: [BEST_EFFORT] 0.82 m (total: 9)
[INFO] [1775641504.283171737] [distance_listener_qos]: [BEST_EFFORT] 0.84 m (total: 10)
[INFO] [1775641504.383216259] [distance_listener_qos]: [BEST_EFFORT] 0.85 m (total: 11)
[INFO] [1775641504.483524358] [distance_listener_qos]: [BEST_EFFORT] 1.18 m (total: 12)
[INFO] [1775641504.583014288] [distance_listener_qos]: [BEST_EFFORT] 1.26 m (total: 13)
[INFO] [1775641504.683119559] [distance_listener_qos]: [BEST_EFFORT] 0.54 m (total: 14)
[INFO] [1775641504.783230864] [distance_listener_qos]: [BEST_EFFORT] 1.19 m (total: 15)
[INFO] [1775641504.883232911] [distance_listener_qos]: [BEST_EFFORT] 1.21 m (total: 16)
[INFO] [1775641504.983324171] [distance_listener_qos]: [BEST_EFFORT] 1.03 m (total: 17)
[INFO] [1775641505.083175280] [distance_listener_qos]: [RELIABLE] 1.26 m (total: 2)
[INFO] [1775641505.083238029] [distance_listener_qos]: [BEST_EFFORT] 1.14 m (total: 18)
[INFO] [1775641505.183414398] [distance_listener_qos]: [BEST_EFFORT] 0.22 m (total: 19)
[INFO] [1775641505.283406917] [distance_listener_qos]: [BEST_EFFORT] 1.30 m (total: 20)
[INFO] [1775641505.383210845] [distance_listener_qos]: [BEST_EFFORT] 0.77 m (total: 21)
[INFO] [1775641505.483263924] [distance_listener_qos]: [BEST_EFFORT] 0.73 m (total: 22)
[INFO] [1775641505.583236877] [distance_listener_qos]: [BEST_EFFORT] 0.20 m (total: 23)
[INFO] [1775641505.683064595] [distance_listener_qos]: [BEST_EFFORT] 0.22 m (total: 24)
[INFO] [1775641505.783291881] [distance_listener_qos]: [BEST_EFFORT] 1.41 m (total: 25)
[INFO] [1775641505.883079478] [distance_listener_qos]: [BEST_EFFORT] 1.01 m (total: 26)
[INFO] [1775641505.982960246] [distance_listener_qos]: [BEST_EFFORT] 1.28 m (total: 27)
[INFO] [1775641506.083579083] [distance_listener_qos]: [RELIABLE] 0.84 m (total: 3)
[INFO] [1775641506.083926295] [distance_listener_qos]: [BEST_EFFORT] 0.53 m (total: 28)
[INFO] [1775641506.182884312] [distance_listener_qos]: [BEST_EFFORT] 0.42 m (total: 29)
[INFO] [1775641506.283128529] [distance_listener_qos]: [BEST_EFFORT] 0.19 m (total: 30)
[INFO] [1775641506.383087300] [distance_listener_qos]: [BEST_EFFORT] 0.80 m (total: 31)
[INFO] [1775641506.483185996] [distance_listener_qos]: [BEST_EFFORT] 0.31 m (total: 32)
[INFO] [1775641506.583334035] [distance_listener_qos]: [BEST_EFFORT] 0.83 m (total: 33)
[INFO] [1775641506.683207598] [distance_listener_qos]: [BEST_EFFORT] 0.51 m (total: 34)
[INFO] [1775641506.783168132] [distance_listener_qos]: [BEST_EFFORT] 0.25 m (total: 35)
[INFO] [1775641506.883379382] [distance_listener_qos]: [BEST_EFFORT] 1.22 m (total: 36)
[INFO] [1775641506.983169217] [distance_listener_qos]: [BEST_EFFORT] 0.18 m (total: 37)
[INFO] [1775641507.083393111] [distance_listener_qos]: [RELIABLE] 1.46 m (total: 4)
[INFO] [1775641507.083945032] [distance_listener_qos]: [BEST_EFFORT] 0.61 m (total: 38)
[INFO] [1775641507.183598749] [distance_listener_qos]: [BEST_EFFORT] 0.81 m (total: 39)
[INFO] [1775641507.283176545] [distance_listener_qos]: [BEST_EFFORT] 1.07 m (total: 40)
[INFO] [1775641507.383420211] [distance_listener_qos]: [BEST_EFFORT] 1.16 m (total: 41)
[INFO] [1775641507.483132792] [distance_listener_qos]: [BEST_EFFORT] 0.20 m (total: 42)
[INFO] [1775641507.583188623] [distance_listener_qos]: [BEST_EFFORT] 1.01 m (total: 43)
[INFO] [1775641507.683140148] [distance_listener_qos]: [BEST_EFFORT] 0.81 m (total: 44)
[INFO] [1775641507.783199381] [distance_listener_qos]: [BEST_EFFORT] 0.99 m (total: 45)
[INFO] [1775641507.883063414] [distance_listener_qos]: [BEST_EFFORT] 0.11 m (total: 46)
[INFO] [1775641507.983257547] [distance_listener_qos]: [BEST_EFFORT] 1.15 m (total: 47)
[INFO] [1775641508.083027772] [distance_listener_qos]: [RELIABLE] 1.41 m (total: 5)
[INFO] [1775641508.083151400] [distance_listener_qos]: [BEST_EFFORT] 0.87 m (total: 48)
[INFO] [1775641508.183647948] [distance_listener_qos]: [BEST_EFFORT] 0.60 m (total: 49)
[INFO] [1775641508.283197985] [distance_listener_qos]: [BEST_EFFORT] 0.96 m (total: 50)
[INFO] [1775641508.312786917] [distance_listener_qos]: — Stats (last 5s) —
[INFO] [1775641508.312895487] [distance_listener_qos]: RELIABLE : 5 msg (expected ~5)
[INFO] [1775641508.312944725] [distance_listener_qos]: BEST_EFFORT : 50 msg (expected ~50)

```

9. QoS INCOMPATIBILITY WARNING APPEARS

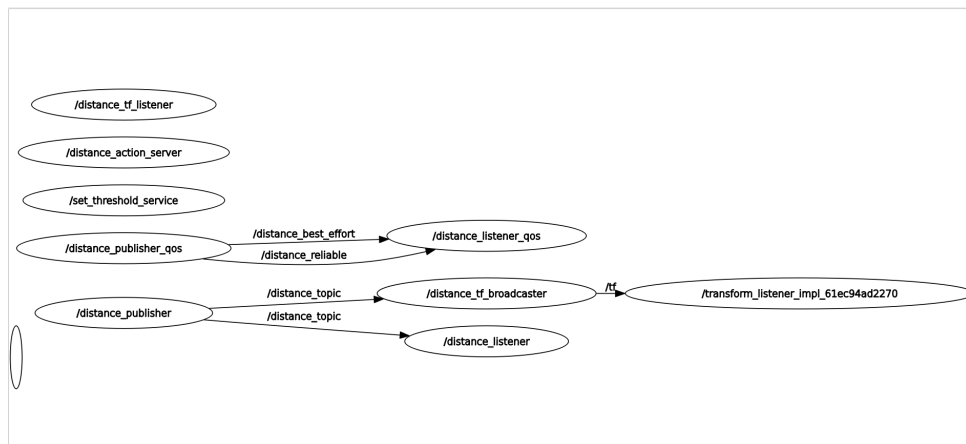
```

c:73
ngxwang@LAPTOP-SS0001C1:~/ros2_ws/src$ ros2 topic echo /distance_best_effort --qos
-reliability reliable
[WARN] [1775641750.362995984] [_ros2cli_11454]: New publisher discovered on topic '
/distance_best_effort', offering incompatible QoS. No messages will be received fro
m it. Last incompatible policy: RELIABILITY

```

rqt_graph

10. FULL LAUNCH FILE STARTS ALL NODES



rqt_graph